

Assessment of Alzheimer's Disease Progression Using Comorbidity-Aware Meta-Bayesian Framework with Resting-State fMRI

Second Review

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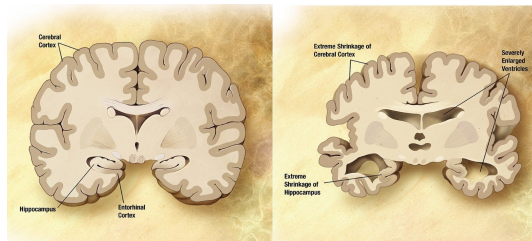
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Academic Year 2026–2027



Background and Motivation

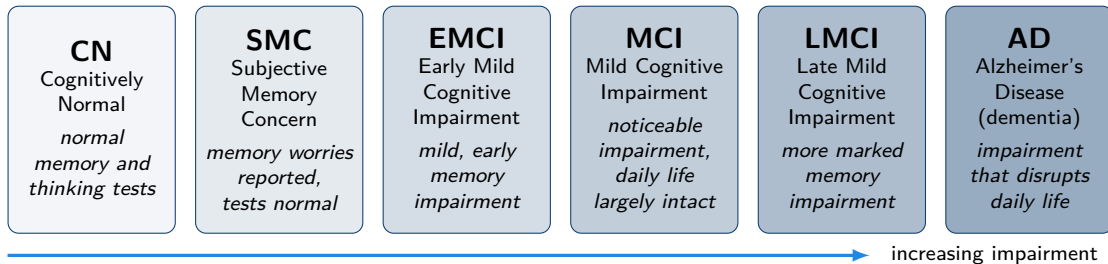
- ▶ **Alzheimer's disease (AD)** is the most common cause of dementia, about 60–70% of cases worldwide.
- ▶ It is **progressive**: memory, reasoning and daily function decline, mostly after age 65.
- ▶ India's ageing population is making the burden grow.
- ▶ Tissue loss (right) appears late. **Brain activity**, measured with functional MRI, can change earlier.



Healthy brain (left) and advanced AD (right): thinner cortex, shrunken hippocampus, enlarged ventricles.

Illustration reused from Review 1.

The Six Stages of Progression



- ▶ Labels are ADNI's diagnostic groups. Each stage has **about 30 scans**.
- ▶ The model treats the stages as an **ordered scale** (fixed at Review I): CN → SMC → EMCI → MCI → LMCI → AD.
- ▶ Neighbouring stages overlap, so every label should come with a **confidence**.

Problem Statement

Problem

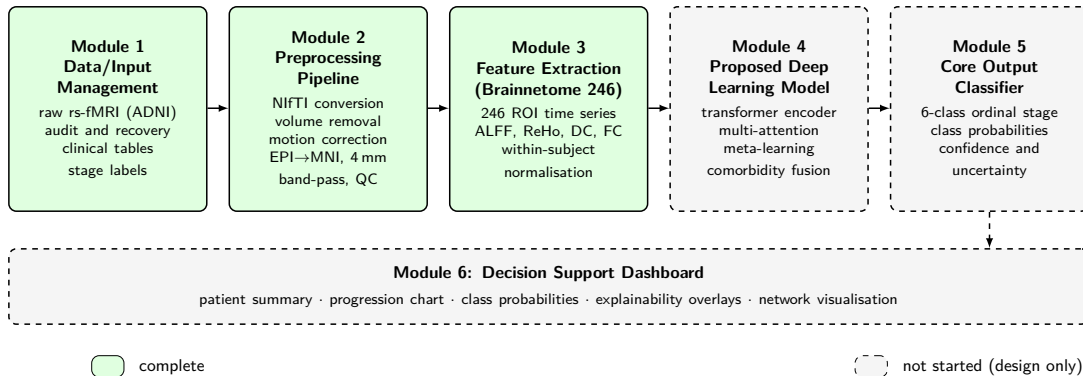
Existing systems analyse rs-fMRI biomarkers **independently** and omit clinical context, giving **overconfident, deterministic staging errors** on borderline subjects.

- ▶ **Context:** dementia progresses through six ordered stages.
- ▶ **Limitation:** pipelines isolate ALFF, ReHo, DC and FC.
- ▶ **Clinical gap:** models ignore how comorbidities modify neurodegeneration.

Research question

Can a common AD progression representation be found across several functional biomarkers, accounting for comorbidity and prediction uncertainty?

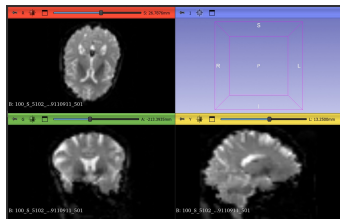
Proposed System Architecture



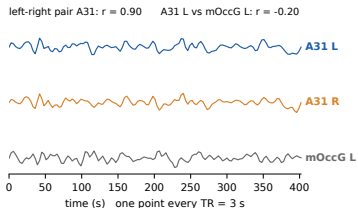
Goal: place each person on the six-stage scale from a resting-state scan and clinical context, and say how sure the model is.

Resting-State fMRI in Plain Words

- ▶ The participant lies still in the scanner and does **no task**.
- ▶ Every 3 s the scanner takes a 3-D picture of the brain: 140 pictures, about 7 minutes.
- ▶ The signal is **BOLD**: local blood oxygenation, a proxy for neural activity.
- ▶ Slow fluctuations (0.01–0.10 Hz) in linked regions rise and fall together.



One converted scan in three planes (Review I).



Three regions from one scan: left and right A31 move together, an occipital region does not.

The Dataset: ADNI

- ▶ **ADNI** is a public, multi-site study that shares brain scans and clinical data [1, 2].
- ▶ We use its **resting-state fMRI only**: the download has no structural (T1) scans.
- ▶ Each scan: 140 volumes, $TR = 3$ s for most, voxels of about 3 mm.
- ▶ 175 scans audited, 2 excluded (1 duplicate, 1 truncated): 173 eligible, **165 processed**. Eight failed: their TR of 6.02 s is too slow for the 0.01–0.10 Hz band.
- ▶ **127 subjects**: some were scanned more than once.

Stage	Elig.	Proc.	Subj.	To 30
CN	29	27	26	+3
SMC	32	31	21	0
EMCI	30	29	26	+1
MCI	35	32	14	0
LMCI	22	21	20	+9
AD	25	25	20	+5
Total	173	165	127	+18

Counts are per acquisition; subjects are people.
Elig.: passed the audit; Proc.: preprocessed; To 30:
processed scans still to add for 30 per stage.

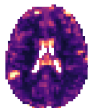
Clinical Context: Covariates and Comorbidity

Covariate	What it is, and why it matters
Age	years at the scan; risk rises with age
APOE ϵ 4	copies of the ϵ 4 variant; strongest common genetic risk factor
MMSE	0–30 cognitive screening score; lower is worse
CDR-SB	0–18 clinician-rated severity; higher is worse
GDS	0–15 depression scale; often accompanies decline
Cardiovascular history	ADNI category MH4CARD; includes hypertension
Endocrine/metabolic history	ADNI category MH9ENDO; includes diabetes

The Four Feature Biomarkers

ALFF

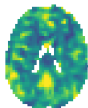
how strong



How big the slow swings of a region's signal are.
Big swings: stronger activity.

ReHo

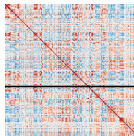
how locally in step



Do the small spots inside a region rise and fall together?
Yes: the region works in step.

FC

who talks to whom



Do two different regions rise and fall together?
The more alike, the more connected.

DC

how connected



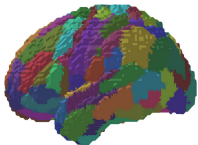
Add up a region's connections to all other regions.
A big total: a hub.

Four simple questions about every brain region

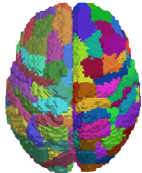
How strongly does it fluctuate? (ALFF) Do its parts move together? (ReHo) Which other regions move with it? (FC) How connected is it overall? (DC)

The Brain Atlas: 246 Regions

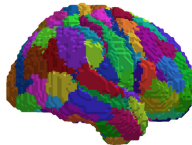
Left hemisphere



Top view



Right hemisphere



Sagittal slice



Coronal slice



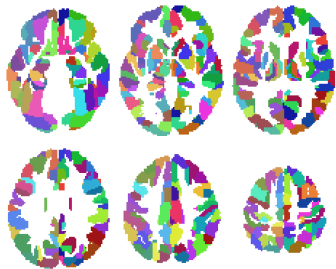
Axial slice



- ▶ An **atlas** divides the brain into named regions so scans can be compared region by region.
- ▶ **Brainnetome atlas**: 246 regions, 123 per hemisphere [3]. Each colour is one region.
- ▶ Every scan is warped to a common template (MNI space), so region k is the same place in every person.
- ▶ A region's signal is the average of its voxels: **246 time series per scan**.

Brainnetome 246-ROI Parcellation for Feature Extraction

Fine-grained, whole-brain representation for systematic regional and inter-regional feature extraction



Axial slices, $z = -16$ to $+56$ mm.

Each colour is one ROI.

- ▶ **Fine-grained whole-brain coverage.** 246 anatomically defined ROIs covering cortical and subcortical regions.
- ▶ **Complete extraction space.** 246/246 ROIs present across all 165 acquisitions; no empty or zero-variance ROIs.
- ▶ **Systematic feature extraction.** Each ROI provides regional biomarkers (ALFF, ReHo, DC) and participates in the 246×246 FC matrix.
- ▶ **No premature feature selection.** All ROIs are retained to preserve spatial information; predictive importance will be learned by the downstream model.

246

ROIs

165

acquisitions

135×246

ROI time series per scan

246×246

FC matrix per scan

The 246 ROIs are not assumed to be equally disease-relevant; they define a complete, reproducible feature-extraction space from which the model will learn predictive regional and connectivity patterns.

What Is New in Our Approach

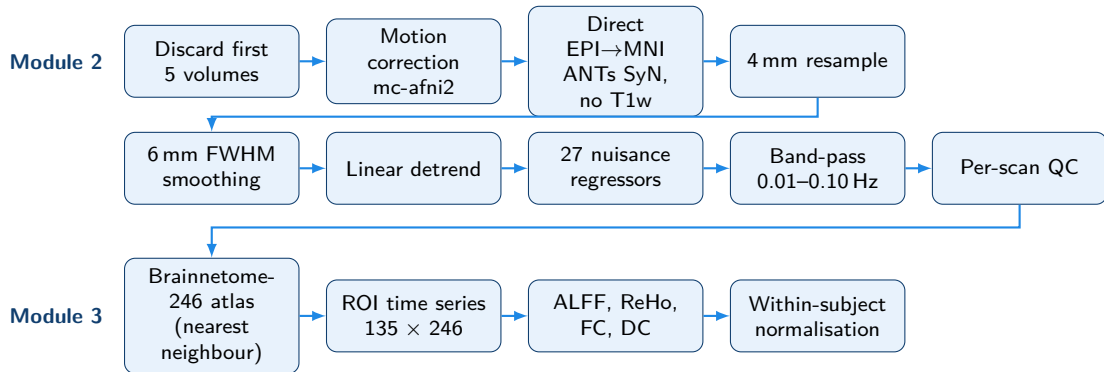
	Gap in existing work	What we propose (design)
G1	Each biomarker is analysed on its own	Meta-learning: each biomarker is a task; learn a shared, biomarker-invariant representation
G2	Clinical context is left out	Fusion: join covariates to the imaging representation, $H = [Z' \parallel C]$
G3	Progression treated as unordered classes	Ordinal model over six ordered stages
G4	Overconfident single labels	Bayesian output: probability per stage with uncertainty

Honest scope

All four are **design only**: nothing is trained. Of three 2026 papers checked, at least one partly addresses each of G2–G4 and none addresses G1 (report §1.3).

System Modules

Module (Review I names)	Status	At Review II
1 Data/Input Management	Complete	Imaging audited; clinical table cleaned (covariates for 107 of 127 subjects)
2 Preprocessing Pipeline	Complete	165 of 173 scans; 8 rejected (TR 6.02 s)
3 Feature Extraction (Brainnetome 246)	Complete	ALFF, ReHo, DC, FC for 165/165; normalised
4 Proposed Deep Learning Model	Not started	Transformer, meta-learning, fusion: design only
5 Core Output Classifier	Not started	Bayesian ordinal head: design only
6 Decision Support Dashboard	Not started	No interface built



Every step is checked per acquisition; a scan that fails a check is logged, never silently altered.

Preprocessing: What Each Step Does and Why

Step	What it does	Why it is needed
Discard first 5 volumes	drops the first 15 s	the signal is still settling at the start
Motion correction	aligns every volume to the first	head movement would otherwise look like brain activity [8, 9]
Warp to MNI template	maps each brain onto a standard brain [10]	so region k is the same place in every person
Resample to 4 mm, smooth 6 mm	coarser grid and slight blur	less noise and fewer voxels
Detrend and 27 regressors	removes slow drift and motion, white-matter and whole-brain signal	non-neural influences hide the neural signal
Band-pass 0.01–0.10 Hz	keeps only slow fluctuations	the range where resting-state activity lives
Quality control	motion, signal-to-noise and file checks per scan	failures are logged, never silently altered

Core Algorithm A: Implemented Biomarkers and Normalisation

Biomarkers (per Brainnetome region)

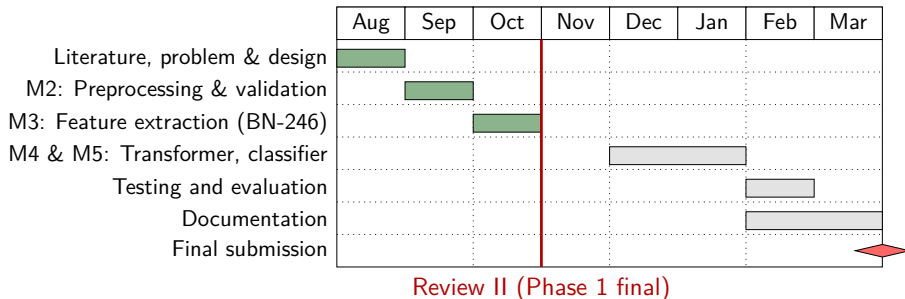
- ▶ **ALFF**: mean FFT amplitude, 0.01–0.10 Hz
- ▶ **ReHo**: Kendall's W , 27-voxel neighbourhood
- ▶ **FC**: 246×246 Pearson, signed, unthresholded
- ▶ **DC**: $DC_i = \sum_{j \neq i} FC_{ij}$

Within-subject normalisation

- ▶ $mALFF = ALFF_i / \overline{ALFF}$
- ▶ $mReHo = ReHo_i / \overline{ReHo}$
- ▶ $DC_z = (DC_i - \overline{DC}) / \sigma_{DC}$
- ▶ $FC \rightarrow \text{artanh}(\text{clip}(r, \pm 0.999999))$, diagonal = 0

- ▶ Every statistic comes from **one acquisition's own 246 values**.
- ▶ No cross-subject statistic and no label: leakage-free by construction.
- ▶ Verified: $\text{mean}(mALFF) = \text{mean}(mReHo) = 1.000000$; $\text{std}(DC_z) = 1.000000$ (165/165).
- ▶ 0 off-diagonal entries needed clipping; Fisher-z diagonal exactly 0.

Implementation Status



One milestone ahead of the Review I plan

Module 2 was planned for Review II and Module 3 for Review III. **Both are complete at Review II**, the final review of Phase 1.

Applies to the **data foundation only** (Modules 1–3); Modules 4–6 are not started. Module 2 completed in September; Module 3 in October (normalised biomarkers 5 Oct).

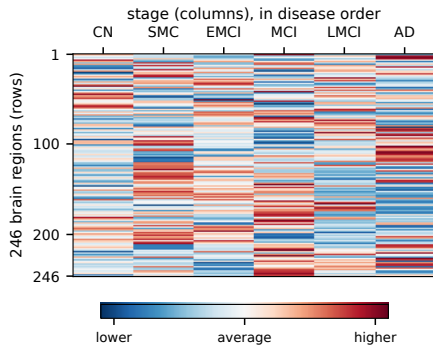
Results: Data Foundation Verification

These are **data-stage** results. **No model performance exists.**

Check (165 acquisitions)	Result
246/246 regions covered	165/165
ROI series (135, 246)	165/165
NaN / Inf / zero-variance	0 / 0 / 0
Pilot reproduction	bit-identical
DC recomputed from FC, max diff.	0.000e+00
Consistency checks passed	165/165

- ▶ 127 subjects, not 165 scans: 96 have one scan, 25 two, 5 three, 1 four.
- ▶ Smallest class: MCI, 32 scans but only **14 subjects**.
- ▶ QC medians ($n = 164$, one artefact scan excluded): tSNR 127.95, SNR 13.12, FD 0.317 mm.

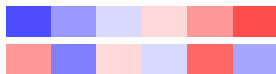
How to Read the Stage Heatmaps



mALFF, 127 subjects (real data).

- ▶ Each **row** is a brain region (246), each **column** a stage. A cell is the group average for that stage.
- ▶ **Colour** is relative to the region's own average over the six stages (row-wise z-score): absolute levels are removed on purpose.
- ▶ Disease-driven change would appear as a row shifting **steadily** from one colour to the other, left to right.
- ▶ **By chance**, an unrelated region is steadily ordered with probability $2/6! \approx 0.28\%$: about **0.7 of 246 regions**. We found 1, 0 and 1.

CN SMC EMCI MCI LMCI AD



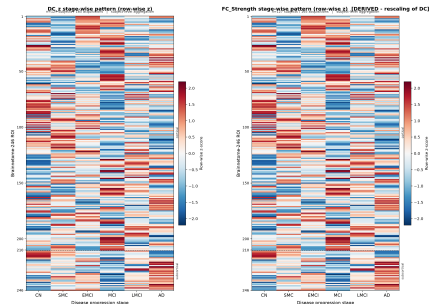
schematic, not data

Finding 1: FC Strength Equals DC/245

Review I fourth column was redundant

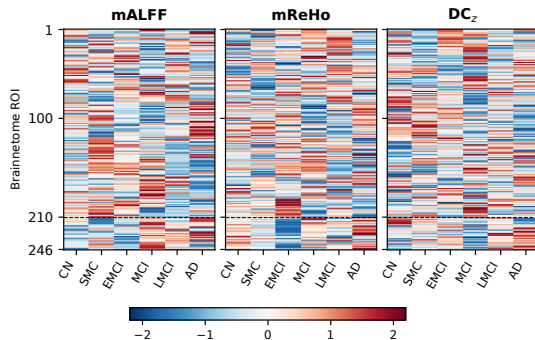
$$\text{FCS}_i = \frac{1}{245} \sum_{j \neq i} \text{FC}_{ij} = \frac{\text{DC}_i}{245}$$

- ▶ Max difference **6.939e-18**; Pearson $r = 1.000000000000$.
- ▶ **All four biomarkers are kept.**
- ▶ ALFF, ReHo, DC: 246×3 regional matrix.
- ▶ FC: full 246×246 Fisher-z matrix.
- ▶ A design correction found by auditing our own features.



DC_z (left) and FC_Strength (right): identical panels, as expected. Descriptive and exploratory only.

Finding 2: No Univariate Progression Signal



Stage-wise row-z patterns, 127 subjects (CN→AD).

Descriptive and exploratory; no statistical test.

	Monotonic CN→AD	By chance
mALFF	1 / 246	≈ 0.7
mReHo	0 / 246	≈ 0.7
DC _z	1 / 246	≈ 0.7

- ▶ No more than chance; stage variation is 4–5× below anatomy.
- ▶ Informative, not a failure.
- ▶ Group means discard the whole connectivity structure.
- ▶ **Motivates the multivariate model.**

Team Responsibilities

Team member	Responsibility	Status at Review II
Harishkanna R	Preprocessing (Module 2); transformer model (Module 4)	Preprocessing: Complete Transformer: Not started
Dhananchezhiyan M	Meta-Bayesian module (meta-learning, Bayesian classifier) and final module (dashboard)	Not started
Nagindhar Sachein R L	Dataset cleaning (clinical and comorbidity tables, dataset expansion)	Clinical cleaning done and audited; dataset expansion next

Feature extraction (Module 3) was completed by the team and is verified.

References

- [1] C. R. Jack Jr. *et al.*, "ADNI: MRI methods," *J. Magn. Reson. Imaging*, 27(4), 2008.
- [2] R. C. Petersen *et al.*, "ADNI: clinical characterization," *Neurology*, 74(3), 2010.
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- [12] R. W. Cox, "AFNI: software for analysis of functional MR neuroimages," *Comput. Biomed. Res.*, 29(3), 1996.
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- [14] C. Finn, P. Abbeel, S. Levine, "Model-agnostic meta-learning," *ICML*, 2017.
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- [17] A. Hariharan, S. N. Gowda, Anala M R, "Uncertainty-aware longitudinal forecasting of AD progression," arXiv:2606.24604, 2026.
- [18] B.-S. Rauchmann *et al.*, "Multimodal ordinal modeling of AD severity (MRI and clinical)," arXiv:2606.11794, 2026.
- [19] K. Alsalem *et al.*, "Progression-aware CNN-Transformer for multiclass AD staging," *Diagnostics*, 16(4):593, 2026.

Thank You

Questions and Discussion

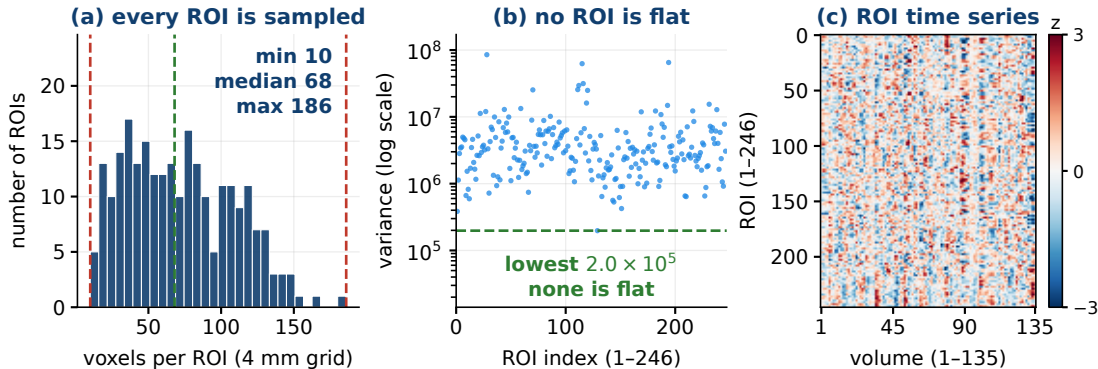
Backup: Glossary

Term	Meaning	Term	Meaning
rs-fMRI	functional MRI recorded while the person rests	ALFF	amplitude of low-frequency fluctuations: size of the slow signal swings
BOLD	blood-oxygen-level-dependent signal measured by fMRI	ReHo	regional homogeneity: how in step a region's voxels are
TR	time between two brain pictures (3 s here)	FC	functional connectivity: correlation of two regions' signals
Voxel	one 3-D pixel of a scan (a 4 mm cube after resampling)	DC	degree centrality: a region's total connectivity (sum of its FC row)
ROI, atlas	region of interest; a map that divides the brain into regions	ADNI	Alzheimer's Disease Neuroimaging Initiative
MNI space	standard template brain that all scans are aligned to	Ordinal	classes with a natural order ($CN < \dots < AD$)
Band-pass	keep only 0.01–0.10 Hz	Meta-learning	learning, across several tasks, what they have in common
Nuisance regression	removing motion and other non-neural signals	Bayesian model	outputs a distribution over predictions, giving uncertainty
Row-wise z	a row rescaled by its own mean and spread	Grouped CV	validation that keeps all scans of one person in the same fold

Backup: Data Limitations

- ▶ **No structural (T1) scans:** each scan was registered directly to the template, with no subject-specific tissue masks, so no CSF regressor: 27 regressors (24 motion terms, white matter, global signal).
- ▶ **Slice-timing correction not applied:** the metadata needed to recover the slice order are not available for this dataset, and nothing was inferred.
- ▶ **Eight scans rejected:** $TR \approx 6.02\text{ s}$ makes the 0.10 Hz band-pass impossible (Nyquist).
- ▶ **No cross-site harmonisation:** ADNI is multi-site, so scanner effects may remain.
- ▶ **127 subjects, 14 in the smallest class (MCI):** model capacity is limited by subjects, not scans; validation must keep each person's scans together.
- ▶ **LMCI covariates missing:** APOE $\epsilon 4$ only for the 20 LMCI subjects; 116 duplicate conflicts in the clinical sources await review.
- ▶ **No univariate stage signal** in the group means (Finding 2): the case for a multivariate model.

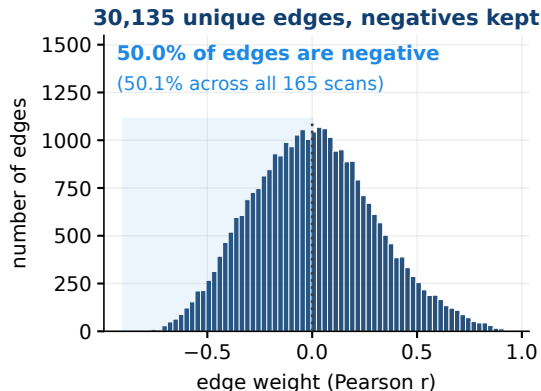
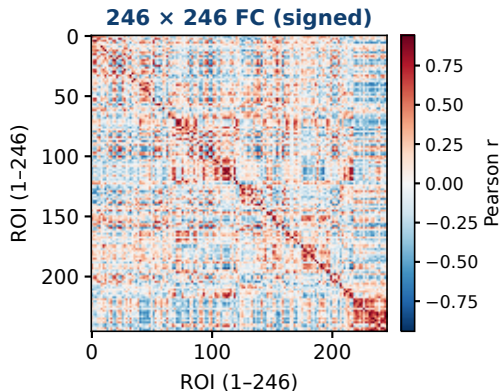
Backup: The 246-ROI Extraction Space Is Verified



246/246 ROIs are present in all 165 acquisitions: no empty ROI (the smallest has 10 voxels), no zero-variance time series, no NaN or Inf. Every scan gives the same 135×246 matrix.

(a): the atlas on the 4 mm grid. (b), (c): one representative CN scan; (c) is z-scored per ROI for display.

Backup: 246 Regions as Connectivity Nodes



Signed and unthresholded: all 30,135 unique edges are kept for each of the 165 acquisitions. About half are negative (50.1 % over the 165 scans; per-scan range 45.6–53.1 %), as expected after global signal regression, which is part of the 27-regressor design and centres the values near zero.

Matrix and histogram: one representative CN scan.

Backup: Technology Stack

Used so far (data foundation)

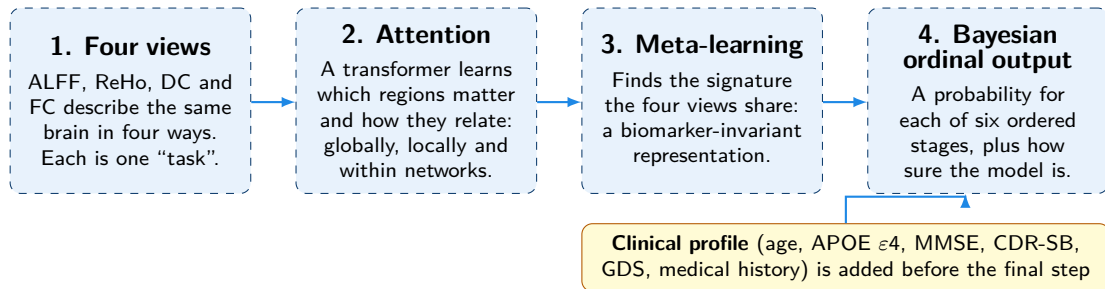
- ▶ Python 3.12.3, NumPy, SciPy, pandas
- ▶ NiBabel 5.4.2, Nilearn 0.14.1
- ▶ ANTsPy 0.6.3 (registration), TemplateFlow 25.1.2
- ▶ FreeSurfer 7.4.1 (FS-FAST motion correction)
- ▶ WSL2 (Ubuntu) on Windows 11; Git

Planned (not yet used)

- ▶ PyTorch, GPyTorch (model, Bayesian layers)
- ▶ Streamlit (Module 6 dashboard)
- ▶ Hardware per Review I: Intel Core i7, 32 GB RAM, RTX 4060 (8 GB)

Dataset: ADNI via LONI. Versions are those recorded in each acquisition's provenance file.

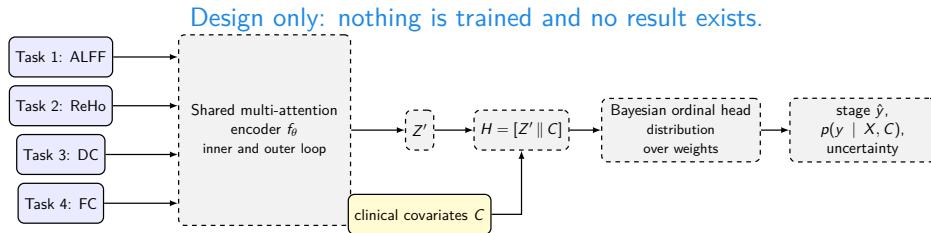
The Meta-Bayesian Model in Plain Words (Design)



- ▶ **Ordinal:** CN→AD is a progression, so confusing neighbouring stages is a smaller error than confusing distant ones.
- ▶ **Bayesian:** borderline subjects get a wide uncertainty instead of a falsely confident label.

Design only. Nothing is trained; no accuracy, calibration or attention result exists.

Core Algorithm B: Meta-Bayesian Pipeline (Design, Not Implemented)



MAML update over biomarkers

$$\theta'_b = \theta - \alpha \nabla_{\theta} \mathcal{L}_b^{\text{sup}}(\theta)$$
$$\theta \leftarrow \theta - \beta \nabla_{\theta} \sum_b \mathcal{L}_b^{\text{qry}}(\theta'_b)$$

Ordinal head (cumulative link)

$$P(y \leq k | H, w) = \sigma(\tau_k - g_w(H)), \quad \tau_0 < \dots < \tau_4$$

Support and query sets are subject-disjoint, within training folds. Uncertainty = predictive entropy over S weight samples.

First Review - Queries and Answers

Panel comment	Response	Detail in
Need more clarity on the dataset (biomarkers)	175 audited, 165 preprocessed, 127 subjects. Four biomarkers on 246 regions.	Dataset and biomarker slides; report §6.1
Need more clarity on the Meta-Bayesian framework	“Meta”: biomarkers as tasks. “Bayesian”: ordinal head with uncertainty. Design only.	Phase 2 slides (at the end); report §4.5
Need more clarity on the comorbidity data	Seven covariates, listed on the covariates slide. Cleaned table for 127 subjects; the LMCI group has APOE only.	Covariates slide; report §4.5